AI-generated · Unofficial practice

Chemistry Mock

Paper 2 (Elective Part)

60 minutes（1 hour）

FOR REVISION

For revision and self-assessment only. Not an official examination paper.

Difficulty rating：**4.0** / 5.0

INSTRUCTIONS

1. This paper consists of THREE sections: Section A Industrial Chemistry, Section B Materials Chemistry, Section C Analytical Chemistry.
2. Attempt ALL questions in any TWO sections.
3. Each section carries 20 marks.
4. Show all working clearly. Unless otherwise specified, give numerical answers to 3 significant figures.
5. A Periodic Table is printed on the last page of this paper.

Independent, unofficial mock paper. Not affiliated with any official examination authority.

Section A Industrial Chemistry

1.

(a) Hydrogen can be manufactured by the following two methods:

Method 1 (steam reforming): $\mathbf{CH_4(g) + 2H_2O(g) \rightarrow CO_2(g) + 4H_2(g)}$

Method 2 (electrolysis of water using renewable electricity): $\mathbf{2H_2O(l) \rightarrow 2H_2(g) + O_2(g)}$

(Relative atomic masses: H = 1.0, C = 12.0, O = 16.0)

(i1) Taking hydrogen as the desired product, compare the atom economies of the two methods and deduce which is higher. (1)

(i2) Even so, Method 2 is considered 'greener'. Give one reason. (1)

(ii1) The first-stage product of steam reforming is syngas. What are the two main gases in syngas? (1)

(ii2) Write the chemical equation for making syngas from methane and steam. (1)

(iii) Hydrogen is also formed in the chloroalkali industry. Apart from hydrogen, name one other product of the chloroalkali industry. (1)

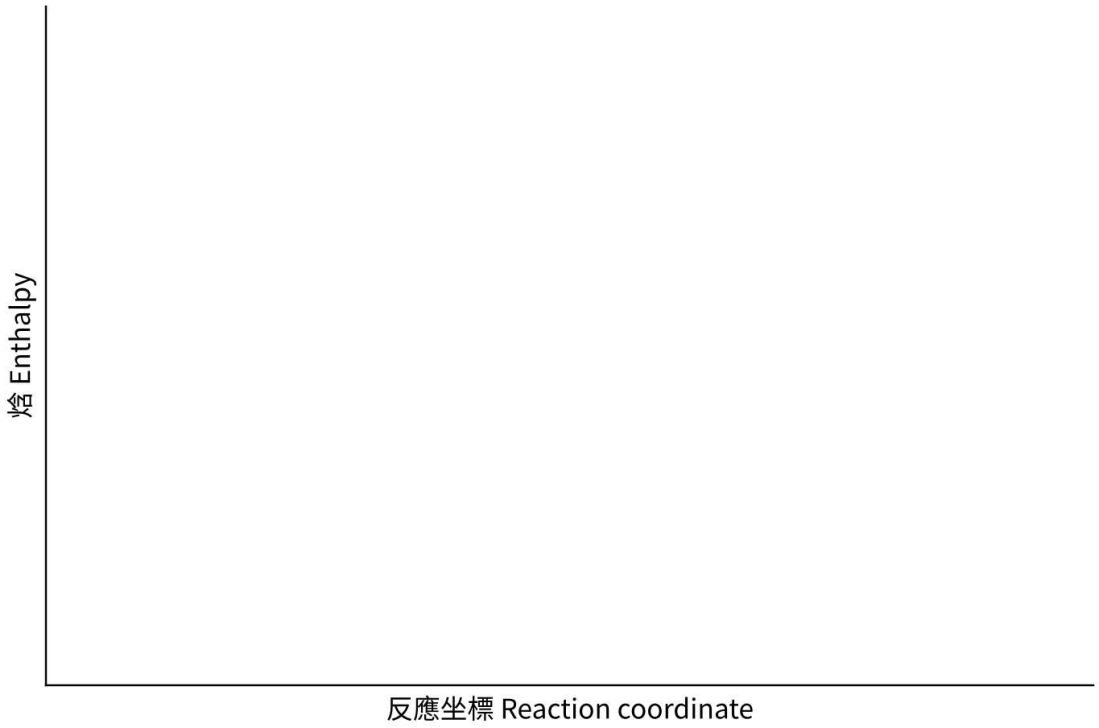
(b) Methanol is made from syngas over a copper-based catalyst:

$\mathbf{CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g) \quad \Delta H = -91\ kJ\ mol^{-1}}$

With the catalyst, the activation energy of the forward reaction is $\mathbf{+80\ kJ\ mol^{-1}}$. The industrial operating conditions are about 250 °C and 70 atm.

(i) What is meant by the term 'activation energy'? (1)

(ii) On the axes below, draw the energy profile of this forward reaction, and label the activation energy and ΔH. (2)



(iii) Write the expression for K_c of the synthesis of methanol, and state how K_c changes when the temperature rises. (1)

(iv) Explain why about 250 °C and 70 atm are chosen in industry. (3)

(v) Explain why the catalyst does not change the equilibrium yield of methanol. (1)

(c) The Deacon process oxidises hydrogen chloride to chlorine. Copper(II) chloride is the catalyst and the process takes place in two steps:

Step 1: $\mathbf{2CuCl_2(s) \rightarrow 2CuCl(s) + Cl_2(g)}$

Step 2: $\mathbf{2CuCl(s) + 2HCl(g) + \frac{1}{2}O_2(g) \rightarrow 2CuCl_2(s) + H_2O(g)}$

(i1) Write the overall equation of this process. (1)

(i2) Explain why only a small amount of copper(II) chloride is needed in this process. (1)

(i3) Explain why water can be used to remove unreacted hydrogen chloride from the product gases. (1)

(ii1) A plant investigates the following reaction at constant temperature: (2)

$2NO(g) + O_2(g) \rightarrow 2NO_2(g)$

Trial 1: $[NO]$ = 0.010 M, $[O_2]$ = 0.010 M, initial rate = 7.0×10^{-6} mol dm^{-3} s^{-1}

Trial 2: 0.020 M, 0.010 M, 2.8×10^{-5}

Trial 3: 0.030 M, 0.020 M, 1.26×10^{-4}

Trial 4: 0.015 M, 0.040 M, initial rate to be found

Deduce the order of reaction with respect to NO and to O_2.

(ii2) Calculate the rate constant, stating its unit, and predict the initial rate in Trial 4. (2)

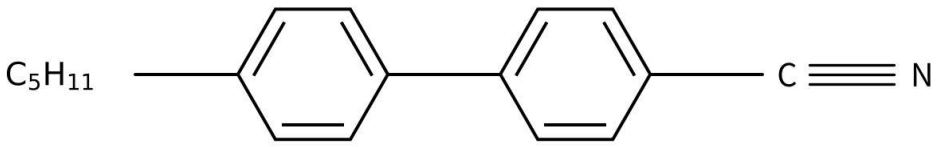
Section B Materials Chemistry

2.

(a) Bio-based polyethene is made by dehydrating ethanol from sugar cane fermentation to ethene and then polymerising it; ordinary polyethene is made from petroleum. The two have the same chemical structure.

(i) Explain why bio-based polyethene is 'greener' than petroleum polyethene in terms of its raw material, and state why it has no such advantage in waste disposal. (2)

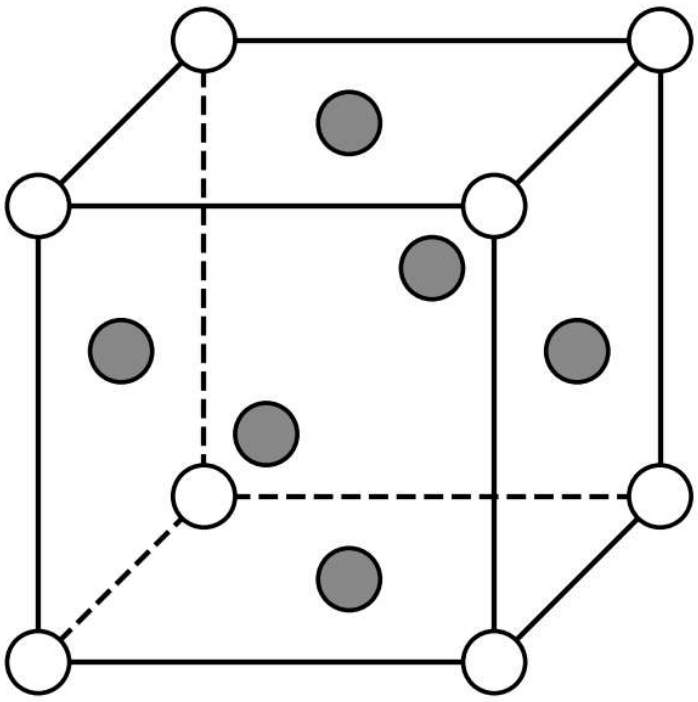
(ii) The diagram below shows a molecule with liquid-crystalline behaviour. State two structural features of this molecule that favour liquid-crystal formation. (2)



(iii) Urea-methanal is used to make the casings of electric plugs. Give one reason. (1)

(b) Copper has a face-centred cubic structure, as shown below.

(i) How many copper atoms are there in a unit cell? (1)



(ii) Based on the data in the diagram, calculate the density of copper, in $\mathbf{g\ cm^{-3}}$. (Relative atomic mass: Cu = 63.5; Avogadro constant = $\mathbf{6.02 \times 10^{23}\ mol^{-1}}$; $\mathbf{1\ nm = 10^{-7}\ cm}$) (2)

(iii1) An aluminium-magnesium alloy is used to make bicycle frames. Explain why the alloy is harder than pure aluminium. (2)

(iii2) The alloy has a lower melting point than pure aluminium. State one industrial advantage of this. (1)

(iv) Making copper into nanoparticles greatly increases its surface area. Compare the same mass of copper made into spherical particles of diameter 10 nm and of diameter 1 000 nm. How many times is the total surface area of the former that of the latter? (2)

(c) Poly(ethylene terephthalate) (PET) is used to make drink bottles.

(i) PET is made from benzene-1,4-dicarboxylic acid and ethane-1,2-diol. State the type of this polymerisation. (1)

(ii) Write the polymerisation equation for the formation of PET. (1)

(iii) Explain why PET is a thermoplastic. (1)

(iv) PET can be converted back to its monomers by hydrolysis (chemical recycling), but polyethene cannot. In terms of their structures, explain this. (2)

(v) Name the moulding method for making PET into continuous sheets. (1)

(vi) Give one reason why PET bottles must be separated from other plastics before recycling. (1)

Section C Analytical Chemistry

3.

(a)(i) Two bottles of white solid have lost their labels. They are potassium carbonate and calcium carbonate. State the expected observations that distinguish them using a flame test. (2)

(ii) Anhydrous sodium carbonate can be used as a primary standard. State one property that makes it suitable. (1)

(iii) Iodine is extracted from iodine solution in water using hexane. In terms of the solubility of iodine and the properties of hexane, explain the principle of this liquid-liquid extraction. (2)

(b) A disinfectant is labelled as containing $30.0\ g\ dm^{-3}$ of hydrogen peroxide. A $10.00\ cm^3$ sample is diluted to $250.0\ cm^3$. $25.00\ cm^3$ of the diluted solution is treated with excess dilute sulphuric acid and titrated directly with $0.0200\ M\ KMnO_4(aq)$. The mean titre is $16.60\ cm^3$.

$$2MnO_4^-(aq) + 5H_2O_2(aq) + 6H^+(aq) \rightarrow 2Mn^{2+}(aq) + 5O_2(g) + 8H_2O(l)$$

(i) Explain why no indicator needs to be added in this titration. (1)

(ii1) State the colour change at the end point. (1)

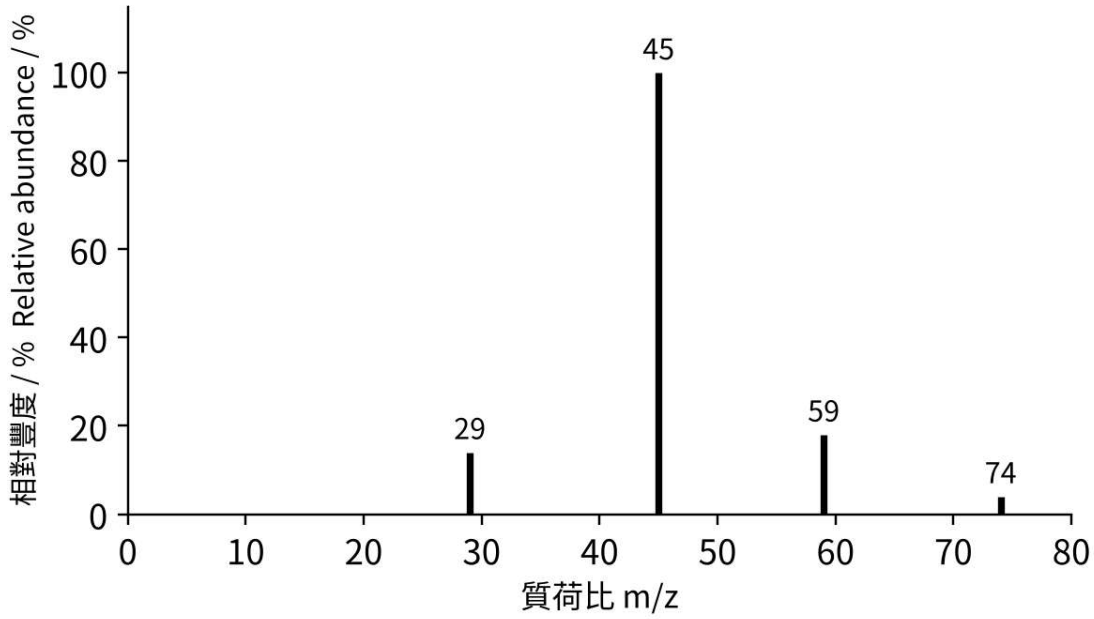
(ii2) Calculate the concentration of hydrogen peroxide in the disinfectant, in $g\ dm^{-3}$. (Relative molecular mass: H_2O_2 = 34.0) (3)

(ii3) The quality control standards require the actual concentration to be within ±5% of the labelled value. By calculation, determine whether this batch of disinfectant passes. (1)

(iii) Explain why sulphuric acid rather than hydrochloric acid is used to acidify the solution. (1)

(c) Liquid compound T is either butan-1-ol or butan-2-ol, with 64.9% C, 13.5% H and 21.6% O by mass. The mass spectrum of T is shown below.

(i1) Deduce the molecular formula of T. (Relative atomic masses: H = 1.0, C = 12.0, O = 16.0) (1)



(i2) With reference to the base peak at m/z 45, and by stating which peak is absent from the diagram, deduce the structural formula of T. (2)

(ii) T is heated with acidified potassium dichromate solution. State the expected observation. (1)

(iii) The organic product V is separated from the reaction in (c)(ii). Write the structural formula of V, and explain why V is not a carboxylic acid. (2)

(iv) Given that the infra-red absorption wavenumber ranges are **1680–1800 cm^{-1}** for C=O and **3230–3670 cm^{-1}** (broad) for O–H in alcohols, explain how the IR spectra of T and V can be compared to show that T has been completely converted. (2)

END OF PAPER

Periodic Table

I	II											III	IV	V	VI	VII	0
1 H 1.0																	2 He 4.0
3 Li 6.9	4 Be 9.0											5 B 10.8	6 C 12.0	7 N 14.0	8 O 16.0	9 F 19.0	10 Ne 20.2
11 Na 23.0	12 Mg 24.3											13 Al 27.0	14 Si 28.1	15 P 31.0	16 S 32.1	17 Cl 35.5	18 Ar 40.0
19 K 39.1	20 Ca 40.1	21 Sc 45.0	22 Ti 47.9	23 V 50.9	24 Cr 52.0	25 Mn 54.9	26 Fe 55.8	27 Co 58.9	28 Ni 58.7	29 Cu 63.5	30 Zn 65.4	31 Ga 69.7	32 Ge 72.6	33 As 74.9	34 Se 79.0	35 Br 79.9	36 Kr 83.8
37 Rb 85.5	38 Sr 87.6	39 Y 88.9	40 Zr 91.2	41 Nb 92.9	42 Mo 95.9	43 Tc (98)	44 Ru 101.1	45 Rh 102.9	46 Pd 106.4	47 Ag 107.9	48 Cd 112.4	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3
55 Cs 132.9	56 Ba 137.3	57 La 138.9	72 Hf 178.5	73 Ta 180.9	74 W 183.9	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.1	79 Au 197.0	80 Hg 200.6	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po (209)	85 At (210)	86 Rn (222)
87 Fr (223)	88 Ra (226)	89 Ac (227)															

atomic number
Symbol
relative atomic mass

Shaded: transition elements. Only the first lanthanoid and actinoid are shown. Values in brackets are mass numbers of the most stable isotopes.

∗Original layout.